

DHARSHAN CS

Backend Software Development Engineer proficient in Python, Flask, REST API design, PostgreSQL, and Redis.

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PROFESSIONAL SUMMARY

Backend engineer who redesigned a browser automation system four times to scale from 1 to 150 concurrent accounts on a single EC2 instance - no infrastructure scaling, purely through architectural improvements. Built a production Flask backend sustaining 300–500 concurrent users at sub-200ms latency. Dual-degree student (IIT Madras + SLIET), with strong foundations in data structures, algorithms, object-oriented design, and scalable, fault-tolerant distributed systems with strong foundations in DSA skills - LC Knight, 1857, top 5%.

EDUCATIONS

Indian Institute of Technology Madras | Chennai, India

Bachelor of Science in Data Science and Applications

Apr 2023

CGPA : 7.36

Sant Longowal Institute of Engineering and Technology | Punjab, India

Bachelor of Engineering in Mechanical Engineering

Nov 2022 - May 2026

CGPA : 7.40

PROJECTS

ParkSpace v2 | 🌐 Github

Backend / Full-Stack Developer

Oct 2025 - Dec 2025

- Architected **RBAC-enabled**, scalable and fault-tolerant backend system sustaining 300–500 concurrent users at sub-200ms latency.
- Engineered 15+ REST API endpoints** with resource-based routing and structured error handling, serving a **Vue.js frontend at sub-200ms** average response time.
- Automated async processing of CSV exports, scheduled emails, and background reports via **Celery + Redis, cutting task latency by 65%** and eliminating blocking ops.
- Cut endpoint **response time from ~850ms to ~210ms via Redis caching** on PostgreSQL dashboard queries, reducing repeated database hits by 40%.
- Modeled a **normalized 6-table relational schema** with referential integrity and dynamic pricing rules, sustaining zero data inconsistency under concurrent load.

Technologies / Tools Used : Python, Flask, PostgreSQL, Redis, Celery, Vue.js

Expertdesk-Monitor | 🌐 Github

Developer

Mar 2024 - Dec 2024

- Identified a manual refresh workflow on an expert Q&A platform and developed a **browser automation system** that eliminated screen monitoring through **Telegram-based real-time alerts**.
- Diagnosed memory constraints across 5 architectural versions, including **Selenium multiprocessing** (OOM at 15 accounts) and **Pyppeteer shared contexts** (memory leak). Redesigned using a **cookie rotation model** to scale to 150 concurrent accounts on a single EC2 instance with stable memory usage.
- Implemented **bidirectional Telegram bot integration** for outbound question alerts and inbound OTP handling, ensuring invisible Unicode character cleaning and character-by-character input to comply with React's keystroke-level validators.
- Developed **automatic session recovery** features, including dead cookie detection, auto re-login, and fresh cookie injection, achieving uninterrupted 24/7 operation.
- Operated as a paid SaaS platform with over **150+ subscribers**.

Technologies / Tools Used : Python, Selenium, Pyppeteer, AWS EC2, Telegram API

Text-to-Learn | AI Course Generator

Backend / Full-Stack Developer

Mar 2026 - May 2026

- Developed a full-stack platform that transforms user prompts into structured courses, featuring modules, lessons, MCQs, and code blocks, utilizing a multi-stage **Gemini API** prompt pipeline.
- Engineered a hierarchical **MongoDB** schema (Course → Module → Lesson) with **Mongoose**, creating over 10 **RESTful** endpoints for efficient CRUD operations within the content hierarchy.
- Created a dynamic lesson renderer in **React** that supports five content block types, including syntax-highlighted code, embedded video queries, and interactive MCQs with answer explanations.
- Secured user-specific routes using **Auth0 OAuth2**, enabling persistent course storage and individual progress tracking across sessions.

Technologies / Tools Used : React, Node.js, Express, MongoDB, Gemini API, Auth0

SKILLS

Computer Science Fundamentals : Data Structures, Algorithms, Object-Oriented Design, Complexity Analysis

Databases : PostgreSQL, SQLite, Redis, MongoDB

Frameworks & Libraries : Flask, Vue.js, Bootstrap, Jinja2, SQLAlchemy, Selenium, Pyppeteer, Celery

Programming Languages : C++, Java, Python, JavaScript, SQL, C

Systems & Backend : Distributed Systems, REST API Design, Async Task Processing, RBAC, Session Management, Cookie Rotation, Load Testing (Locust)

Tools & Platforms : Git, Linux, AWS EC2, Telegram API

AWARDS & ACHIEVEMENTS

- LeetCode Rating: 1859 - Top 5% globally (500,000+ users)**
- Ranked 245th** out of **30,000+** participants in LeetCode Weekly Contest 497